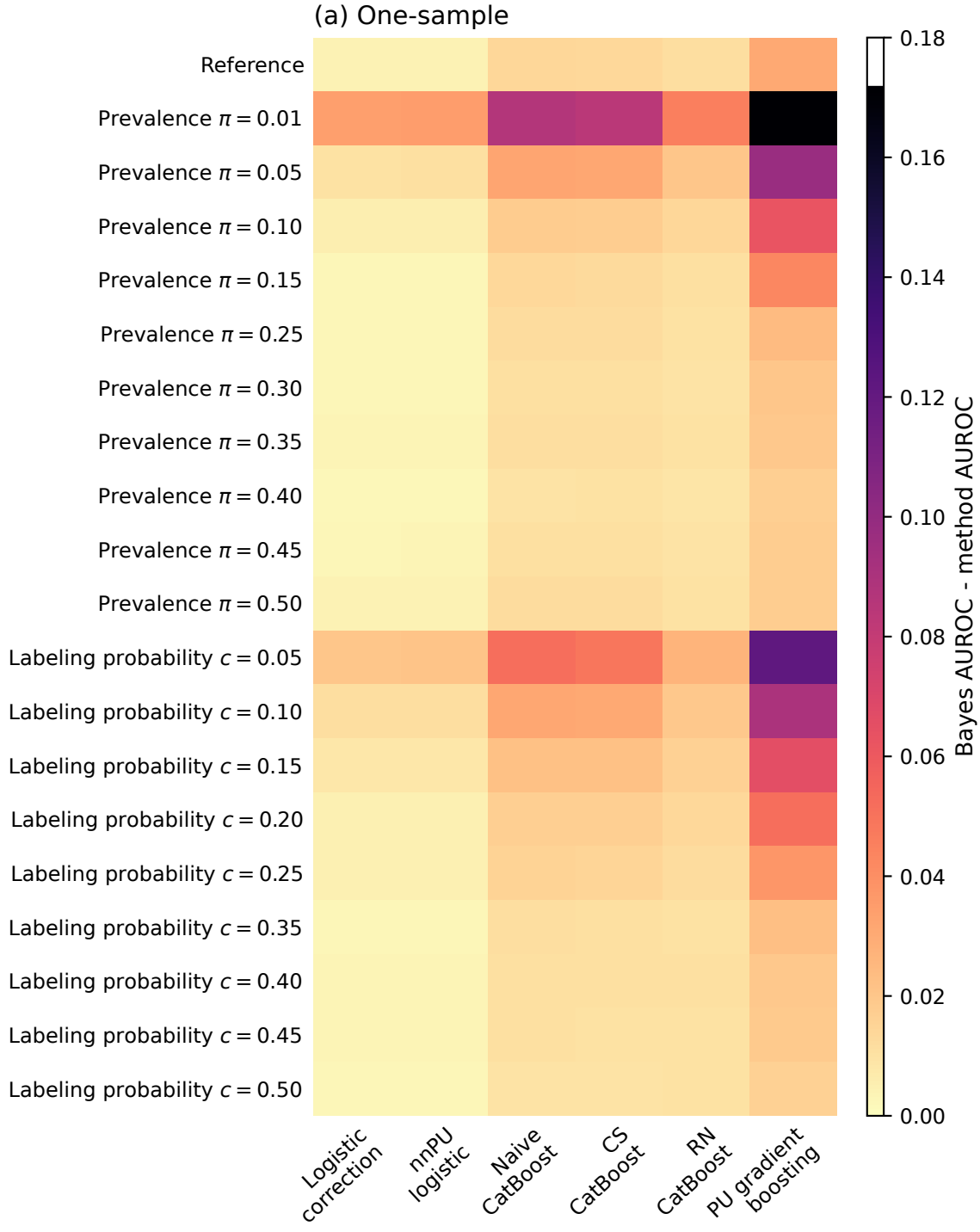
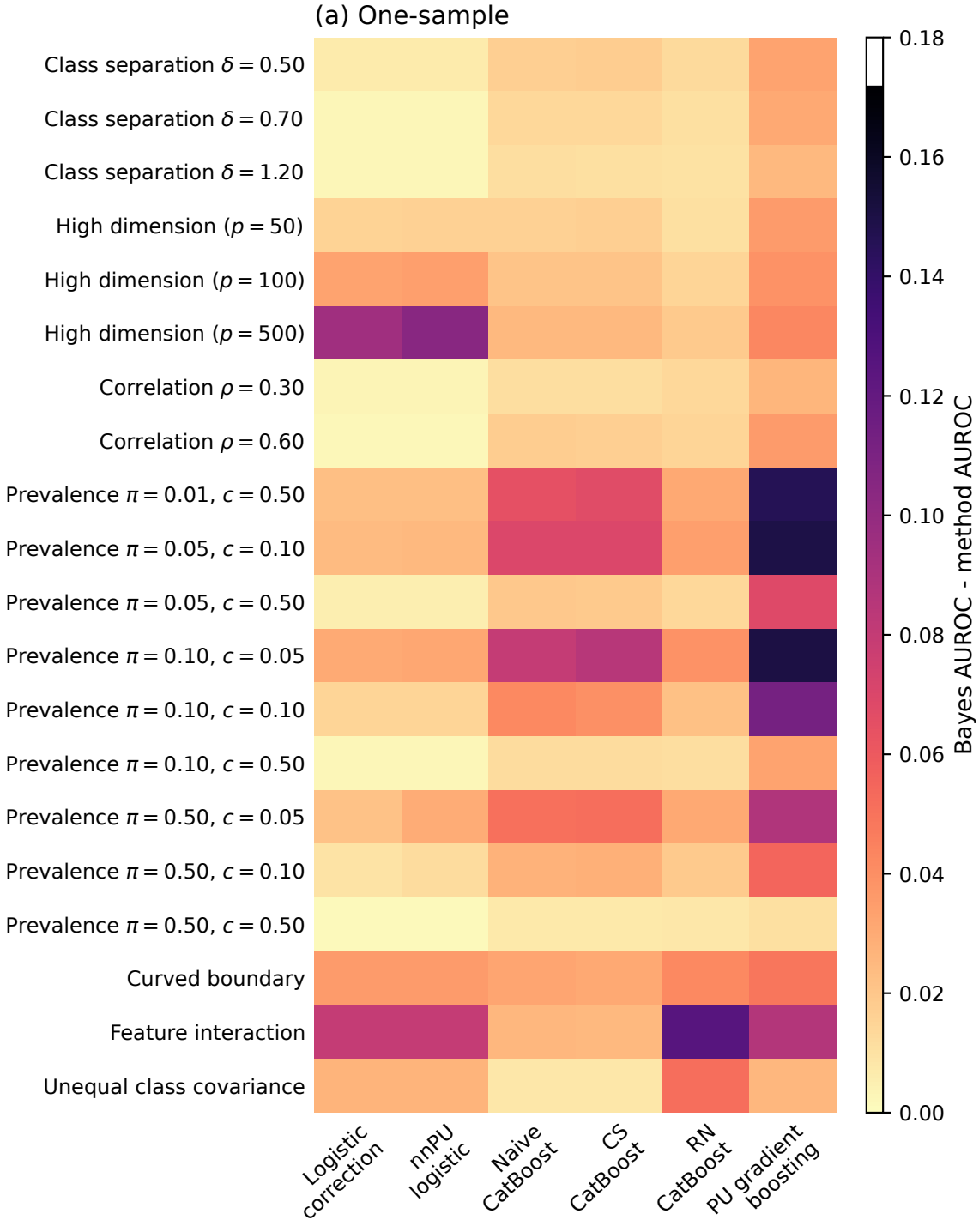
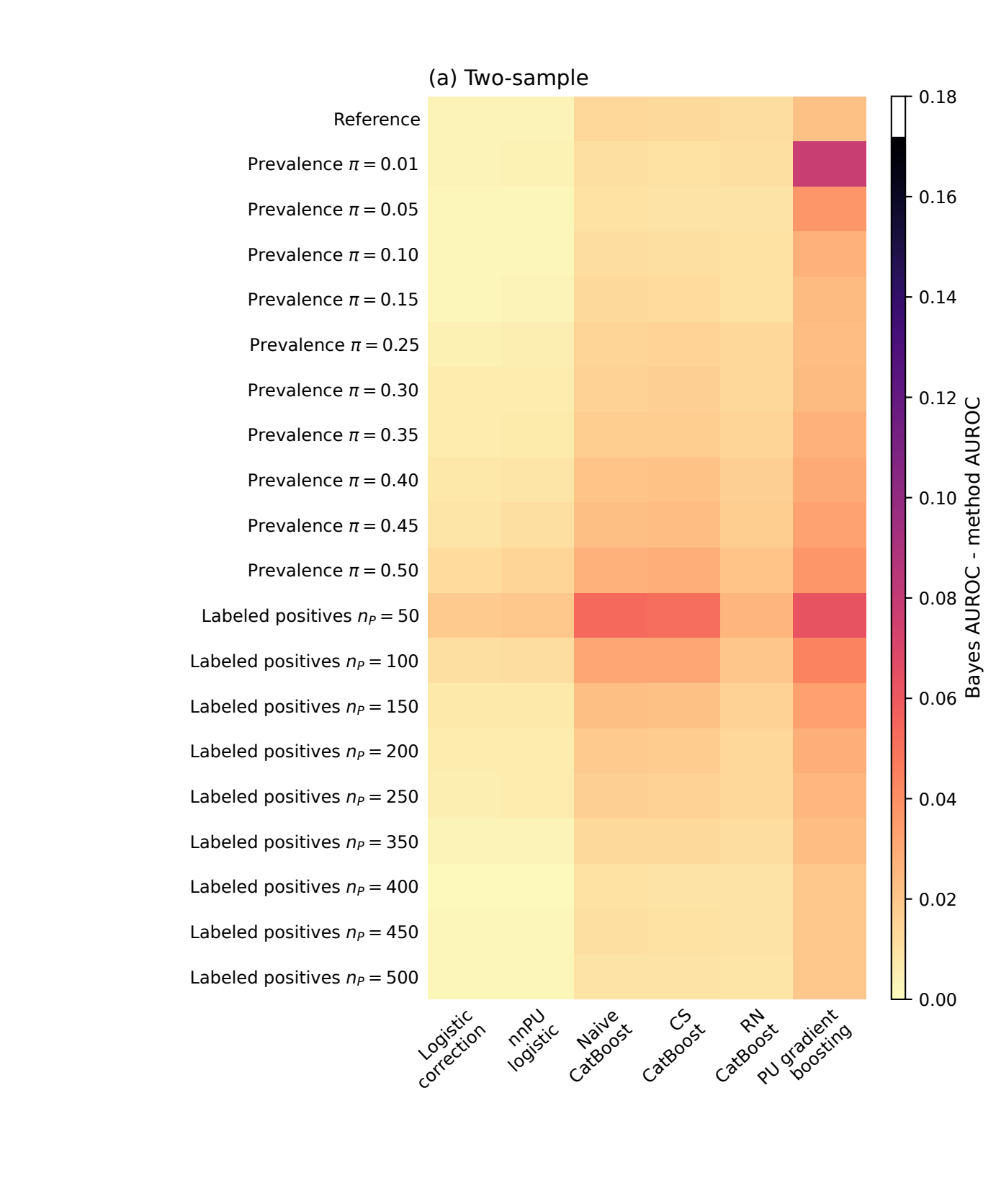
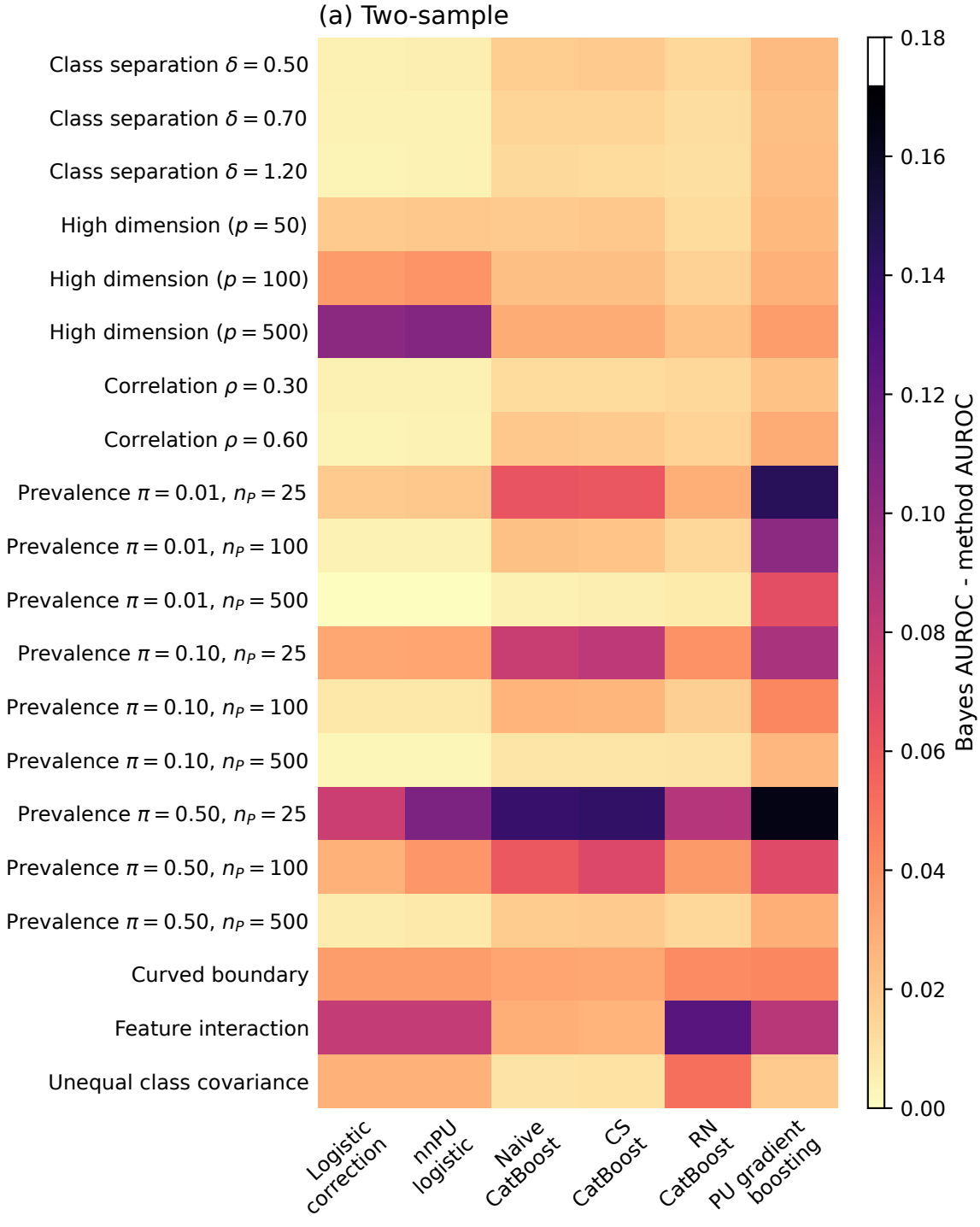

(a) One-sample
Reference
Prevalence $\pi = 0.01$
Prevalence $\pi = 0.05$
Prevalence $\pi = 0.10$
Prevalence $\pi = 0.15$
Prevalence $\pi = 0.25$
Prevalence $\pi = 0.30$
Prevalence $\pi = 0.35$
Prevalence $\pi = 0.40$
Prevalence $\pi = 0.45$
Prevalence $\pi = 0.50$
Labeling probability $c = 0.05$
Labeling probability $c = 0.10$
Labeling probability $c = 0.15$
Labeling probability $c = 0.20$
Labeling probability $c = 0.25$
Labeling probability $c = 0.35$
Labeling probability $c = 0.40$
Labeling probability $c = 0.45$
Labeling probability $c = 0.50$
Logistic correction
nnPU logistic
Naive CatBoost
CS CatBoost
RN CatBoost
PU gradient boosting
Bayes AUROC - method AUROC
0.18
0.16
0.14
0.12
0.10
0.08
0.06
0.04
0.02
0.00


(a) One-sample
Class separation $\delta = 0.50$
Class separation $\delta = 0.70$
Class separation $\delta = 1.20$
High dimension ($p = 50$)
High dimension ($p = 100$)
High dimension ($p = 500$)
Correlation $\rho = 0.30$
Correlation $\rho = 0.60$
Prevalence $\pi = 0.01$, $c = 0.50$
Prevalence $\pi = 0.05$, $c = 0.10$
Prevalence $\pi = 0.05$, $c = 0.50$
Prevalence $\pi = 0.10$, $c = 0.05$
Prevalence $\pi = 0.10$, $c = 0.10$
Prevalence $\pi = 0.10$, $c = 0.50$
Prevalence $\pi = 0.50$, $c = 0.05$
Prevalence $\pi = 0.50$, $c = 0.10$
Prevalence $\pi = 0.50$, $c = 0.50$
Curved boundary
Feature interaction
Unequal class covariance
Logistic correction
nnPU logistic
Naive CatBoost
CS CatBoost
RN CatBoost
PU gradient boosting
Bayes AUROC - method AUROC
0.18
0.16
0.14
0.12
0.10
0.08
0.06
0.04
0.02
0.00


(a) Two-sample
Reference
Prevalence $\pi = 0.01$
Prevalence $\pi = 0.05$
Prevalence $\pi = 0.10$
Prevalence $\pi = 0.15$
Prevalence $\pi = 0.25$
Prevalence $\pi = 0.30$
Prevalence $\pi = 0.35$
Prevalence $\pi = 0.40$
Prevalence $\pi = 0.45$
Prevalence $\pi = 0.50$
Labeled positives $n_P = 50$
Labeled positives $n_P = 100$
Labeled positives $n_P = 150$
Labeled positives $n_P = 200$
Labeled positives $n_P = 250$
Labeled positives $n_P = 350$
Labeled positives $n_P = 400$
Labeled positives $n_P = 450$
Labeled positives $n_P = 500$
Logistic correction
nnPU logistic
Naive CatBoost
CS CatBoost
RN CatBoost
PU gradient boosting
Bayes AUROC - method AUROC
0.18
0.16
0.14
0.12
0.10
0.08
0.06
0.04
0.02
0.00


(a) Two-sample
Class separation $\delta = 0.50$
Class separation $\delta = 0.70$
Class separation $\delta = 1.20$
High dimension ($p = 50$)
High dimension ($p = 100$)
High dimension ($p = 500$)
Correlation $\rho = 0.30$
Correlation $\rho = 0.60$
Prevalence $\pi = 0.01$, $n_P = 25$
Prevalence $\pi = 0.01$, $n_P = 100$
Prevalence $\pi = 0.01$, $n_P = 500$
Prevalence $\pi = 0.10$, $n_P = 25$
Prevalence $\pi = 0.10$, $n_P = 100$
Prevalence $\pi = 0.10$, $n_P = 500$
Prevalence $\pi = 0.50$, $n_P = 25$
Prevalence $\pi = 0.50$, $n_P = 100$
Prevalence $\pi = 0.50$, $n_P = 500$
Curved boundary
Feature interaction
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